INCH-POUND

A-A-52048B
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SUPERSEDING
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3 May 1999

COMMERCIAL ITEM DESCRIPTION

THERMOMETER, SELF-INDICATING, LIQUID-IN-GLASS, POCKET TYPE

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

1.1 Abstract. This commercial item description covers a thermometer, graduated in degrees Fahrenheit, for the determination of atmospheric temperatures during surveying operations.

2. SALIENT CHARACTERISTICS.

2.1 Thermometer. The thermometer (see figure 1) shall be as specified herein and shall be provided with a protective case when specified (see 6.2).

2.1.1 Stem. The stem shall be a magnifying type with a broad white or other contrasting strip at the back to facilitate accurate reading of the thermometer. The stem shall be annealed, and formed to maximize the angle at which it can be read and the apparent width of the thermosensitive fluid column. The top of the stem shall be a button configuration, formed to fit into and be retained by the case. To ensure interface with existing cases, the thermometer shall not exceed the dimensions shown in figure 1.

2.1.2 Bulb. The bulb shall be cylindrical in shape with its diameter shall be less than that of the stem. The bulb, as the reservoir of the thermosensitive fluid shall be sufficient to accommodate the fluid column adjusting to changing temperatures.

Comments, suggestions, or questions on this document should be addressed to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN: CEERD-TS-T, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864 or emailed to pd-te@tec.army.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil .

AMSC N/A FSC 6685

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2.1.3 Graduations. The thermometer shall be capable of measuring temperatures between -66 °F and 160 °F, calibrated in two degree increments. Each 10 degree increment shall be distinctively marked, and each 20 degree increment shall be identified, for ease of temperature determination. All marks and numbers shall be indelible, black for zero and above and in red for below zero. The letter “F”, in black, shall be inscribed at the upper limit of the scale graduations (see figure 1).

2.1.4 Thermosensitive fluid. The thermosensitive fluid shall be of sufficient quantity to indicate the temperature, in the range specified, in any ambient temperature and encased to ensure an accuracy of ±3 °F. The fluid used shall minimize any tendency of the liquid column to separate. The stem and the thermosensitive fluid shall be free of gas bubbles or other foreign material.

2.2 Case. The case shall be rigid, unbreakable, with a non-reflective outer surface, and shall securely hold the thermometer centered inside the case. The case shall have a cap that secures firmly to the case and seals against the entrance of moisture or other foreign material. The cap shall retain the stem top (button) for easy withdrawal of the thermometer for use. A ball-pointed, shirt pocket type clip shall be affixed to the outside of the case, close to the cap end, to permit secure retention of the case and contents when carried.

3. REGULATORY REQUIREMENTS.

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The unit provided shall meet the salient characteristics of this CID, and conform to the manufacturer’s drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The unit offered shall be the latest model of the standard product of the supplier that has been used in the field, by commercial organizations, for at least one year.

4.3 Examination. Each thermometer shall be examined for compliance with the requirements specified in Section 2. This inspection shall encompass all examinations of performance and dimensional requirements. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. Failure to comply with any specified requirement or the presence of one or more defects shall constitute cause for rejection.

5. PACKAGING.

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

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6. NOTES.

6.1 Intended use. This thermometer is intended for use in determining media temperatures required for use in surveying operations.

6.2 Ordering data.

a. Title, number, and date of this document.
b. When a case is required (see 2.1).
c. Packaging (see 5.1).

6.3 Data requirements. The contracting officer shall include, in the contract or order, any data requirements, such as instructional materials and technical documentation to be furnished with each thermometer.

6.4 National Stock Numbers (NSNs). The following NSN is assigned to a product that conforms to this CID. This may not be indicative of all possible NSNs associated with the CID.

NSN
6685-00-174-6235

6.5 Subject term (key word) listing.

Dry bulb
Surveying
Thermosensitive

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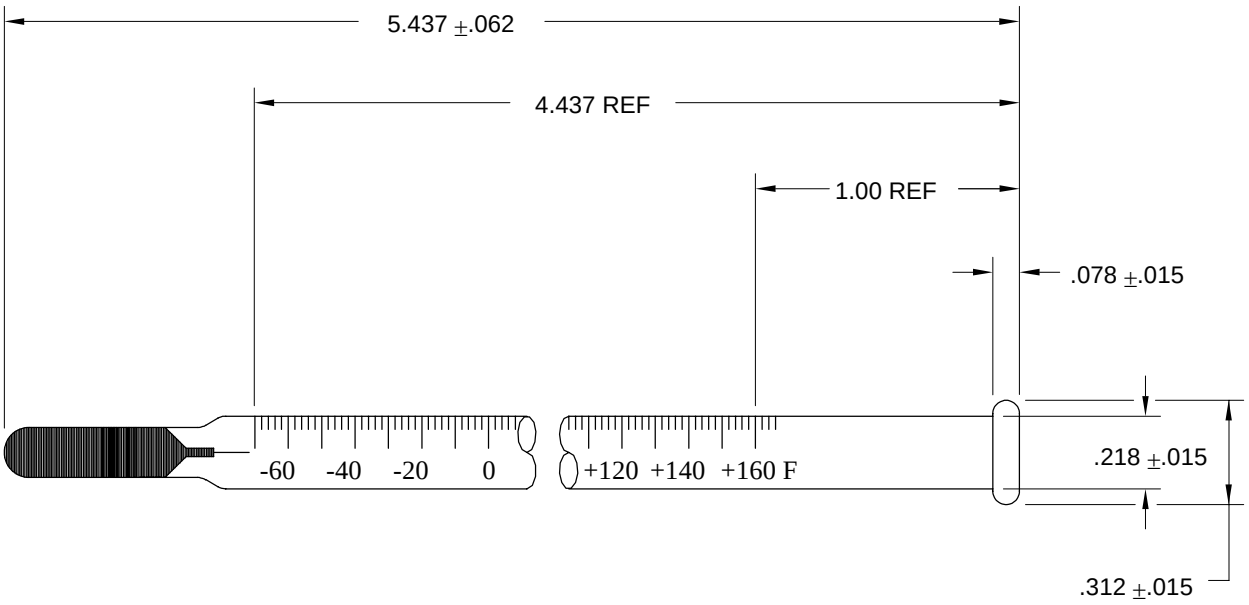

DIMENSIONS ARE IN INCHES.

Figure 1. Thermometer - Example

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